EJERCICIO 1:

$R = \rho \frac{l}{A}$; $R_1 = \rho \frac{l_1}{A_1} = 6\,\Omega$

$l_2 = 3 l_1$; $A_1 l_1 = A_2 l_2 = 3 A_2 l_1 \longrightarrow A_2 = A_1/3$

$R_2 = \rho \frac{l_2}{A_2} = \rho \frac{3 l_1}{A_1/3} = 9 \rho \frac{l_1}{A_1} = 9 R_1 = 9 \times 6\,\Omega = \boxed{54\,\Omega}$

EJERCICIO 2: $R_{20} = \rho_{20} \frac{l}{A}$; $\rho_{20} = 1{,}7 \times 10^{-8}\,\Omega m$

$A = 1\,mm^2 = 10^{-6}\,m^2$

$l = 5000 \times 91\,m \longrightarrow l = 500\,m$

$R_{20} = \frac{1{,}7 \times 10^{-8}\,\Omega m \times 5000\,m}{10^{-6}\,m^2} \longrightarrow R_{20} = 8{,}5\,\Omega$

$R(T) = \frac{l}{A} R(T) = \underbrace{\frac{l}{A} \rho_{20}}_{= R_{20}} \left[1 + \alpha (T - 20^\circ C)\right]$

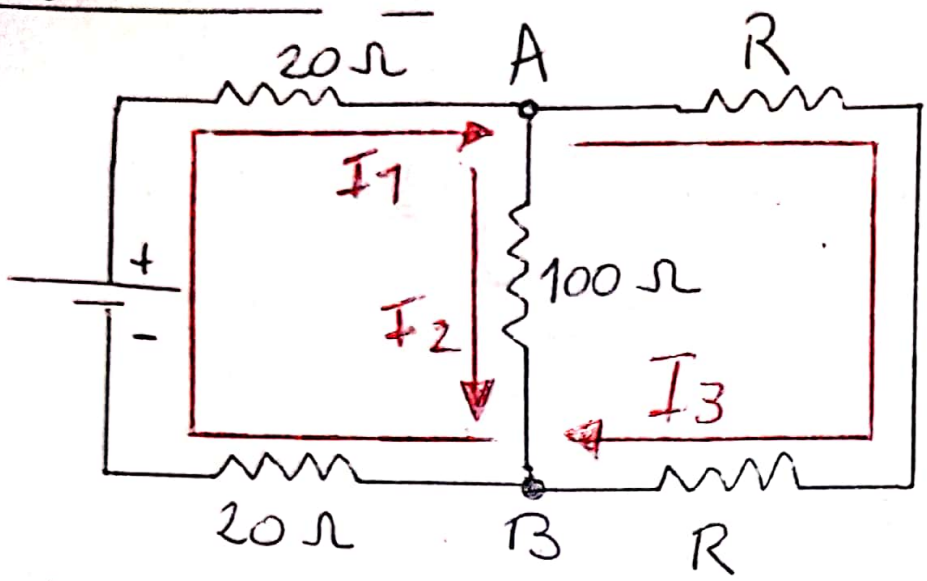
$\longrightarrow R(T) = R_{20}\left[1 + \alpha (T - 20)\right] \quad \left(\alpha = 0{,}0039\ \frac{1}{^\circ C}\right)$

$10{,}2\,\Omega = 8{,}5\,\Omega \left[1 + 0{,}0039\ \frac{1}{^\circ C} (T - 20^\circ C)\right]$

$\longrightarrow \boxed{\Delta T = 51{,}28^\circ C \ (T = 71{,}28^\circ C)}$

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EJERCICIO 8



a) $0{,}45\ W = 20\ \Omega\ I_1^2 \longrightarrow I_1 = 0{,}15\ A$

$0{,}25\ W = 100\ \Omega\ I_2^2 \longrightarrow I_2 = 0{,}05\ A$

$I_1 = I_2 + I_3 \longrightarrow I_3 = 0{,}1\ A$

$V_{AB} = 100\ \Omega\ I_2 = 5\ V = 2R\ I_3 \rightarrow \boxed{R = 25\ \Omega}$

b) $P_R = R I_3^2 = 25\ \Omega \times (0{,}1\ A)^2 = \boxed{0{,}25\ W}$

c) $P_F = P_{DIS} = (20\ \Omega\ I_1^2) \times 2 + 100\ \Omega\ I_2^2 +$

$+ (25\ \Omega\ I_3^2) \times 2$

$\boxed{P_F = 1{,}65\ W}$

d) $V I_1 = P_F \longrightarrow V = \frac{1{,}65\ W}{0{,}15\ A} \longrightarrow \boxed{V = 11\ V}$

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EJERCICIO 20

2) $R_1 = \frac{l}{\sigma_1 A}$ $R_2 = \frac{l}{\sigma_2 A} = \frac{l}{2\sigma_1 A} = \frac{R_1}{2}$ FALSO

σ MAYOR → RESISTENCIA MENOR //

3) $\frac{1}{R_{//}} = \frac{1}{R_1} + \frac{1}{R_2} \rightarrow \frac{1}{R_{//}} > \frac{1}{R_1} \rightarrow R_{//} < R_1$

$R_{//} < R_2$

FALSO

4) $R_S = 3R$

$\frac{1}{R_{//}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R} \rightarrow R_{//} = \frac{R}{3}$

VERDADERO

5) SERIE → MISMA I POR AMBAS

$P_{V_1} + P_{V_2} = IV_1 + IV_2 = I(V_1 + V_2) = P_{V_1+V_2}$

FALSO

1) Y 6) FALSAS

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